

Is Spike-Driven Self-Attention Necessary? The Inefficiency of Jaccard Attention in Spiking Sentence Embeddings

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Abstract

Spiking Neural Networks (SNNs) provide a highly energy-efficient alternative to conventional Deep Neural Networks by utilizing discrete, event-driven temporal spikes. However, adopting Transformer-based paradigms into the spiking domain typically requires mapping complex Self-Attention mechanisms into spike operations. In this paper, we present an in-depth mathematical formulation and empirical ablation of a Spike-Driven Self-Attention layer—modeled around a Jaccard-like spike intersection metric—to evaluate its true utility in semantic sentence embedding tasks. Through comprehensive knowledge distillation, we demonstrate that incorporating Jaccard Attention yields an execution bottleneck (latency increase of $\sim 2.08\times$) and tripling the total spike emission, while offering a negligible absolute accuracy gain (+0.0048 Pearson correlation). We propose a purely recurrent, attention-free Spiking Sentence Embedder operating exclusively through Leaky Integrate-and-Fire (LIF) pooling and Backpropagation Through Time (BPTT). Our minimalist architecture achieves a highly competitive Pearson correlation of 0.8031 on the ALL-STS benchmark with an estimated energy savings ratio of $1298\times$ compared to standard Transformer MACs, conclusively showing that spatial routing via self-attention is redundant for high-fidelity spiking semantic representations.

1 Introduction

The computational overhead of Dense Sentence Embedders (e.g., SBERT, SimCSE) is heavily dominated by the dense matrix multiplications required in the standard Scaled Dot-Product Attention. Spiking Neural Networks (SNNs) alleviate this constraint by replacing Multiply-Accumulate (MAC) operations with sparse Synaptic Operations (SOPs). Recent research has focused heavily on translating Self-Attention to SNNs. However, the fundamental question persists: *Does the temporal recurrent dynamics of an SNN naturally obviate the need for explicit spatial attention mechanisms?*

In this study, we systematically deconstruct the forward dynamics and BPTT learning mechanisms of an SNN sentence embedder. We mathematically define a Jaccard-based Spiking Attention algorithm and contrast it against an attention-free Recurrent Pooler. Our empirical findings across a multi-phase evaluation indicate that the temporal integration capabilities of the pooler are overwhelmingly sufficient for encoding semantic structures.

Our contributions are threefold:

1. **Dynamic SNN Pipeline:** We present a scalable knowledge distillation pipeline optimized for spiking models, dynamically scaling embedding dimensions up to $d_{model} = 256$.
2. **Empirical Ablation:** We evaluate a Jaccard-based Spike Attention mechanism against a purely Recurrent Spiking Pooler, demonstrating that spatial attention adds severe overhead without commensurate accuracy improvements.
3. **Energy Efficiency Analysis:** We report a state-of-the-art energy efficiency formulation, achieving over $1000\times$ theoretical energy savings compared to conventional Transformers while maintaining ~ 0.80 Pearson correlation on the Semantic Textual Similarity (STS) benchmark.

2 Forward Dynamics and Spiking Architecture

Our framework consists of three principal stages: Tokenization, Embedding Spike Generation, and Recurrent Integration.

2.1 Spiking Embedding Layer

Raw text is tokenized using Byte-Pair Encoding (BPE) into a sequence of vocabulary indices. Let $X \in \mathbb{R}^{L \times d_{vocab}}$ be the one-hot encoded input sequence of length L . The Spiking Embedding Layer translates these tokens into temporal spike trains over discrete time steps $t \in [1, T]$.

The membrane potential of the embedding layer V_{emb} is modeled via Leaky Integrate-and-Fire (LIF) dynamics:

$$V_{emb}(t) = \beta_{emb} V_{emb}(t-1) + XW_{emb} \quad (1)$$

A spike $S_{emb}(t) \in \{0, 1\}^{L \times d_{model}}$ is emitted when the potential exceeds a trainable threshold θ_{emb} :

$$S_{emb}(t) = \Theta(V_{emb}(t) - \theta_{emb}) \quad (2)$$

where $\Theta(\cdot)$ is the Heaviside step function. Upon spiking, a soft-reset is implicitly applied by subtracting θ_{emb} .

2.2 Spike-Driven Jaccard Attention (Ablation Variant)

To evaluate the impact of attention, we formalize a Spike-Driven Self-Attention mechanism. Unlike floating-point dot products, our method computes similarities through the intersection of active spikes (equivalent to the numerator of the Jaccard index).

The embedding spikes $S_{emb}(t)$ are projected into Query (Q), Key (K), and Value (V) spikes using independent LIF neurons:

$$Q(t) = \Theta(\beta_Q V_Q(t-1) + S_{emb}(t)W_Q - \theta_Q) \quad (3)$$

$$K(t) = \Theta(\beta_K V_K(t-1) + S_{emb}(t)W_K - \theta_K) \quad (4)$$

$$V(t) = \Theta(\beta_V V_V(t-1) + S_{emb}(t)W_V - \theta_V) \quad (5)$$

For any query token i and key token j , the raw attention score is derived by counting the simultaneous spike overlaps across the d_{model} feature dimensions:

$$\text{MatchCount}_{i,j}(t) = \sum_{d=1}^{d_{model}} Q_{i,d}(t) \wedge K_{j,d}(t) \quad (6)$$

$$\text{Score}_{i,j}(t) = \frac{\text{MatchCount}_{i,j}(t)}{\max_k (\text{MatchCount}_{i,k}(t) + \epsilon)} \quad (7)$$

These normalized scores drive a separate LIF neuron to generate attention mask spikes $S_{score}(t)$, which gate the Value spikes to form the aggregated output $A(t)$:

$$A_i(t) = \sum_{j=1}^L S_{score_{i,j}}(t) \cdot V_j(t) \quad (8)$$

While purely spike-driven, this calculation demands $\mathcal{O}(L^2 \cdot d_{model})$ Boolean operations per timestep T , severely hampering execution speeds.

2.3 The Recurrent Spiking Pooler (Proposed)

In our proposed attention-free architecture, $S_{emb}(t)$ bypasses attention and directly feeds into the Recurrent Pooler. The pooler integrates the token-level spikes spatially and temporally to produce a single dense semantic vector representation $E \in \mathbb{R}^{d_{model}}$.

The pooler’s membrane potential V_{pool} evolves as:

$$V_{pool}(t) = \beta_{pool}V_{pool}(t-1) + S_{emb}(t)W_{pool} + B_{pool} \quad (9)$$

$$S_{pool}(t) = \Theta(V_{pool}(t) - \theta_{pool}) \quad (10)$$

The final continuous sentence embedding E is extracted by accumulating the membrane potentials across the entire temporal sequence T :

$$E = \sum_{t=1}^T V_{pool}(t) \quad (11)$$

This accumulation allows the network to encode nuanced syntactic dependencies purely through temporal state decay β_{pool} and recurrent temporal accumulation, completely bypassing the $\mathcal{O}(L^2)$ spatial routing matrix.

3 Learning Dynamics: Surrogate Gradients and BPTT

The entire pipeline is trained end-to-end via Knowledge Distillation from a continuous Teacher Transformer model. The objective is to minimize the Mean Squared Error (MSE) between the Cosine Similarity of the SNN embeddings and the Teacher’s score.

$$\mathcal{L} = \left(\frac{E_1 \cdot E_2}{\|E_1\| \|E_2\|} - y_{teacher} \right)^2 \quad (12)$$

3.1 Surrogate Gradient Approximation

Because the Heaviside step function $\Theta(x)$ has a derivative of zero almost everywhere, we approximate its gradient $\sigma'(V)$ using a rectangular window surrogate function:

$$\frac{\partial S(t)}{\partial V(t)} \approx \sigma'(V(t)) = \begin{cases} 1 & \text{if } |V(t) - \theta| < \text{window_size} \\ 0 & \text{otherwise} \end{cases} \quad (13)$$

3.2 Backpropagation Through Time (BPTT)

Learning occurs by unrolling the network over time T . For the Recurrent Pooler, the error signal at time t , denoted $\delta(t)$, receives gradients from the immediate output layer and from the future time step $t+1$:

$$\delta(t) = \frac{\partial \mathcal{L}}{\partial V(t)} + (\delta(t+1) \cdot \beta_{pool} \cdot \sigma'(V(t))) \quad (14)$$

Using the Add-Only Delta rule, the synaptic weights W_{pool} are updated exclusively using the pre-synaptic spikes $S_{emb}(t)$ and the masked temporal error $\tilde{\delta}(t) = \delta(t) \cdot \sigma'(V(t))$:

$$\Delta W_{pool} = \eta \sum_{t=1}^T S_{emb}(t)^\top \tilde{\delta}(t) \quad (15)$$

This explicit time-dependency formulation ensures that the pooler learns to map long-term contextual semantic dependencies naturally, offsetting the lack of a self-attention mechanism.

4 Experimental Setup

We evaluate our models using a diverse set of Semantic Textual Similarity (STS) benchmarks. The models are implemented in Rust to allow extreme control over memory execution and matrix operations, guaranteeing bit-accurate MAC-to-SOP translation.

Datasets: The evaluation spans STS-12 through STS-16, STS-B, SICK-R, and an aggregated ALL-STs set encompassing 29,720 testing pairs.

Metric: Model quality is measured via the Pearson Correlation coefficient of cosine similarities against gold standard scores. Efficiency is evaluated via Inference Latency (ms/pair) and Spiking Synaptic Operations (SOPs).

5 Results and Discussion

5.1 Phase 1: Training Strategy Efficacy

We first determine the optimal training methodology by comparing standard Contrastive Learning (SimCSE), Supervised Human-Only scores, and Knowledge Distillation from a Teacher Model. All models here were evaluated at a baseline dimension $d_{model} = 64$.

Table 1: Phase 1: Training Strategy Efficacy ($d_{model} = 64$)

Training Strategy	Train Time (s)	ALL-STs Pearson	STS-B Pearson
SimCSE (Unsupervised)	470.6	0.6477	0.5849
Human-Only (Supervised)	7.1	0.6665	0.6089
Knowledge Distillation	1074.7	0.7091	0.6656

Discussion: Knowledge Distillation massively outperforms both unsupervised SimCSE and supervised learning from sparse human labels, demonstrating that SNNs highly benefit from dense, continuous signal regression.

5.2 Phase 2: Dimensionality Scaling

Using the Distillation strategy, we scaled the embedding dimension d_{model} to find the optimal balance between representation capacity and efficiency.

Table 2: Phase 2: Dimensionality Scaling

d_{model}	ALL-STs Pearson	Inference Latency (ms/pair)	Est. SOPs/ Sentence	Energy Savings
32	0.6713	0.718	17,284	82,991×
64	0.7091	0.911	79,505	18,042×
128	0.8038	1.341	339,181	4,229×
256	0.8079	1.946	1,407,442	1,019×

Discussion: The SNN scales extremely well up to 128 and 256 dimensions, breaching the 0.80 Pearson correlation barrier, which is highly competitive with continuous full-precision models, while still retaining massive energy savings relative to 1.4 G-MAC Transformer models.

5.3 Phase 3: Ablation of Spike-Driven Attention

Fixing the dimension to $d_{model} = 256$, we perform a structural ablation against our Spike-Driven Jaccard Attention and the minimalist No-Attention (Recurrent Pooler only). We also evaluated a variant using Homogeneous Initializations.

Table 3: Phase 3: Ablation of Spike-Driven Attention ($d_{model} = 256$)

Architecture Variant	ALL-STS Pearson	Inference (ms/pair)	Avg Spikes/ Sent.	Energy Savings
No-Attention (Proposed)	0.8031	0.932	2,157	1298×
Jaccard Attention	0.8079	1.946	6,737	1019×
Homogeneous-Init (Attn)	0.7996	1.924	6,687	1179×

Discussion: Our empirical results definitively undermine the necessity of the Spike-Driven Self-Attention block. While the Jaccard Attention variant slightly edges out in raw accuracy ($\Delta = +0.0048$ Pearson), it emits over $3\times$ the amount of internal spikes and requires twice the execution time (1.946 ms vs 0.932 ms) per pair.

Mathematically, the capacity of the Recurrent Pooler’s BPTT gradients to propagate temporal context $\beta V(t-1)$ proves highly expressive. The baseline No-Attention model achieves striking parity while maintaining maximum synaptic sparsity and execution speed.

6 Conclusion

Through detailed mathematical formulation and rigorous empirical distillation, we demonstrate that embedding Spatial Attention (Jaccard Spike Overlap) into Spiking Neural Networks creates an unjustifiable computational bottleneck. The temporal decay properties inherent to LIF neurons, when trained optimally via BPTT, naturally encapsulate complex sentence semantics without the need for quadratic sequence routing. We conclude that Spike-Driven Self-Attention is redundant for dense semantic representations, paving the way for ultra-efficient, purely recurrent SNN embedders.